

ABAHA MONDAL

M.Sc. in Big Data Analytics

Ramakrishna Mission Vivekananda Educational and Research Institute, Belur Math, West Bengal, India

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PROJECTS

- **CleanVision using DIV2K:** Deep Learning based Real-Time Image Denoising and Restoration
 - Built a real-time denoising system using DnCNN and CNN architectures to handle Gaussian, Speckle, and Salt & Pepper noise.
 - Optimized performance with PSNR and SSIM metrics, including hyperparameter sensitivity analysis.
 - Deployed as an interactive web app using Gradio for real-time testing.
- **Rainfall Prediction Analysis:** Machine Learning for Rainfall Forecasting
 - Built rainfall prediction models using Logistic Regression, Decision Tree, Random Forest, SVM, XGBoost, and KNN.
 - Evaluated models with Accuracy, F1-score, and ROC-AUC; XGBoost achieved the best performance.
 - Conducted end-to-end implementation on Kaggle using Python, NumPy, and Pandas.
- **Automated Short Answer Grading (ASAG):** NLP-based LLM Project for Multi-Modal DocQA
 - Implemented a short-answer grading system using the MDocAgent framework with multi-modal and multi-agent components.
 - Integrated OCR, retrieval-augmented generation (RAG), and visual embeddings for context-aware evaluation.
 - Improved grading accuracy on diverse responses using specialized agents and Hugging Face datasets.
- **Movie Recommender System:** Built and deployed a recommender system for movies focusing mainly on Content-Based and Collaborative Filtering
 - Data preparation and feature engineering from raw movie metadata.
 - Content-based recommendation using text vectorization and similarity computation.
 - Streamlit-based frontend development using PyTorch and deployment as a web application.

COURSEWORK

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|--------------------------------|------------------------------------|
| • Data Structures & Algorithms | • Computer Vision |
| • Database Management System | • Time Series Analysis |
| • Machine Learning | • Java & Hadoop |
| • Deep Learning & NLP | • Probability & Stochastic process |
| • Reinforcement Learning & AI | • Advanced Statistics |

ACHIEVEMENTS & CERTIFICATIONS

- **SVMCM Scholarship** – Awarded for academic excellence during B.Sc.
- **CUET PG Qualified** – Mathematics & Data Science, conducted by NTA
- **NPTEL Certification** – The Joy of Computing Using Python

EDUCATION

- **Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI), Howrah**
M.Sc. in Big Data Analytics
2024 – Present CGPA: 6.09
- **AJC Bose College, University of Calcutta, Kolkata**
B.Sc. (Hons.) in Mathematics
2020 – 2024 CGPA: 7.97
- **St. Paul's Mission School, Kolkata**
Indian School Certificate (ISC), Class XII
2018 – 2020 Score: 87.5%
- **St. Paul's Mission School, Kolkata**
Indian Certificate of Secondary Education (ICSE), Class X
2007 – 2018 Score: 90.08%

TECHNICAL SKILLS

- **Programming Languages:** Python, C, Java, R, $\LaTeX$
- **Frameworks & Libraries:** PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
- **Tools & Platforms:** Git/GitHub, AWS, Hadoop, Oracle Database, Power BI, Google Analytics, MS Office
- **Operating Systems:** Windows, Linux (Ubuntu)

ACTIVITY

- **Chegg (Remote) – Subject Matter Expert (Mathematics)** Provided detailed, step-by-step solutions to advanced mathematics problems for global learners.
- **Fest Organiser – Perceptron 2025 (Chess Group)** Core committee member responsible for planning and managing chess events during the annual tech fest.
- **Problem Solving (HackerRank & LeetCode)** Practiced Python and Data Structures & Algorithms through competitive programming challenges.

HOBBY

- Cricket
- Music
- Chess
- Photography